Chapter 7: Linearly Solvable MDPs

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Abstract. *In this chapter, we study a class of stochastic optimal control problems commonly known as* KL control *problems. The name "KL" came from the quantity called Kullback-Leibler divergence, which serves as the cost functional to be minimized in this class of control problems. The purpose of this chapter is to formulate discrete-time discrete-state KL control problems and to discuss algorithmic advantages they offer (both conceptually and computationally). Specifically, we will show that under the "deterministic transition property" the KL control problems can be solved by a backward dynamic programming algorithm which only involves simple linear operations. This explains why KL control problems are also known as* linearly solvable Markov decision problems *[1]. We will also show that under the same assumption, the KL control problems can be solved using Monte-Carlo simulations.*

Contents

7.1 Problem formulation: Discrete-time, discrete-state case

7.1.1 State transition model

Let $\mathcal{T} = \{0, 1, ..., T\}$ be the set of discrete time index. For each $t \in \mathcal{T}$, suppose a finite state space $\mathcal{X}_t = \{1, 2, ..., |\mathcal{X}_t|\}$ and a finite set of control actions $\mathcal{U}_t = \{1, 2, ..., |\mathcal{U}_t|\}$ are given. In general, we allow both $\mathcal{X}_t$ and $\mathcal{U}_t$ to be time-varying. For each $t \in \{0, 1, ..., T-1\}$, let

$$P_{X_{t+1}|X_t,U_t}(x_{t+1}|x_t, u_t) = \Pr\{X_{t+1} = x_{t+1}|X_t = x_t, U_t = u_t\} \tag{7.1}$$

be the conditional probability distribution representing the state transition probability. When there is no danger of confusion, $P_{X_{t+1}|X_t,U_t}(x_{t+1}|x_t, u_t)$ will be simply written as $P(x_{t+1}|x_t, u_t)$. For each $t \in \{0, 1, ..., T-1\}$, let

$$Q_{U_t|X_t}(u_t|x_t) = \Pr\{U_t = u_t|X_t = x_t\} \tag{7.2}$$

be the conditional probability distribution representing the control policy. In the KL control problem, our goal is to design the control policy $\{Q_{U_t|X_t}\}_{t=0}^{T-1}$ to minimize the cost functional we will introduce shortly in Section 7.1.2 below. In general, we allow the control policy $Q_{U_t|X_t}$ to be randomized – that is, under the optimal policy, U_t is not a deterministic function of X_t.

Assuming that the initial state distribution $P_{X_0}(x_0)$ is given and a control policy $\{Q_{U_t|X_t}\}_{t=0}^{T-1}$ is fixed, the joint probability distribution representing the distribution of the closed-loop trajectories is uniquely determined by

$$\begin{aligned} &Q_{X_{0:T},U_{0:T-1}}(x_{0:T}, u_{0:T-1}) \\ &= \prod_{t=0}^{T-1} P_{X_{t+1}|X_t,U_t}(x_{t+1}|x_t, u_t) Q_{U_t|X_t}(u_t|x_t) P_{X_0}(x_0). \end{aligned} \tag{7.3}$$

Here, we used the symbol "Q" to denote the distribution of the closed-loop trajectories to indicate that it is induced by the control policy $\{Q_{U_t|X_t}\}_{t=0}^{T-1}$. If a different control policy $\{R_{U_t|X_t}\}_{t=0}^{T-1}$ is used, we will write the induced joint distribution by

$$\begin{aligned} &R_{X_{0:T},U_{0:T-1}}(x_{0:T}, u_{0:T-1}) \\ &= \prod_{t=0}^{T-1} P_{X_{t+1}|X_t,U_t}(x_{t+1}|x_t, u_t) R_{U_t|X_t}(u_t|x_t) P_{X_0}(x_0). \end{aligned} \tag{7.4}$$

For each $t \in \mathcal{T}$, the state distribution $Q_{X_t}(x_t)$ induced by the policy $\{Q_{U_t|X_t}\}_{t=0}^{T-1}$ is obtained by marginalizing (7.3). Alternatively, it can be computed recursively by

$$Q_{X_{t+1}}(x_{t+1}) = \sum_{x_t \in \mathcal{X}_t} \sum_{u_t \in \mathcal{U}_t} P_{X_{t+1}|X_t,U_t}(x_{t+1}|x_t, u_t) Q_{U_t|X_t}(u_t|x_t) Q_{X_t}(x_t) \tag{7.5}$$

with $Q_{X_0}(x_0) = P_{X_0}(x_0)$.

7.1.2 Cost functional

The cost functional to be minimized is a stage-additive cost functional comprised of the following two terms:

State-dependent cost

Let $C_t(\cdot,\cdot):\mathcal{X}_t\times\mathcal{U}_t\to\mathbb{R}$ be a given real-valued function. The quantity $C_t(x_t,u_t)$ can be thought of as the cost of a pair of random variables (X_t,U_t) taking a realization (x_t,u_t). The state dependent cost at time step t is the expectation[1]

$$\mathbb{E}^Q C_t(X_t,U_t)=\sum_{x_t\in\mathcal{X}_t}\sum_{u_t\in\mathcal{U}_t} Q_{X_t}(x_t)Q_{U_t|X_t}(u_t|x_t)C_t(x_t,u_t). \tag{7.6}$$

KL-divergence cost

Recall that the relative entropy (KL-divergence) from a probability distribution R on $\mathcal{X}$ to another probability distribution Q on $\mathcal{X}$ is defined as

$$D(Q\|R)=\sum_{x\in\mathcal{X}} Q(x)\log\frac{Q(x)}{R(x)}. \tag{7.7}$$

We assume the use of natural logarithms throughout this note. We also use the convention that $0\log 0=0$, $a\log\frac{a}{0}=\infty$ if $a>0$ and $0\log\frac{0}{0}=0$. It is well-known that $D(Q\|R)$ is a distance-like function in the sense that $D(Q\|R)\geq 0$ for all pairs (Q,R) where the equality holds if and only if $P=Q$. However, the relationship is asymmetric in that $D(Q\|R)\neq D(R\|Q)$ in general.

In the KL control problems, we assume that a "nominal" or a "reference" control policy $\{R_{U_t|X_t}\}_{t=0,1,\ldots,T-1}$ is given as part of the problem formulation, and the deviation of the synthesized policy $Q_{U_t|X_t}$ from $R_{U_t|X_t}$ is penalized using the KL divergence. That is, the stage-wise deviation cost is the expected value of the KL divergence:

$$\begin{aligned}&\mathbb{E}^Q D(Q_{U_t|X_t}(\cdot|X_t)\|R_{U_t|X_t}(\cdot|X_t))\\ &=\sum_{x_t\in\mathcal{X}_t}\sum_{u_t\in\mathcal{U}_t} Q_{X_t}(x_t)Q_{U_t|X_t}(u_t|x_t)\log\frac{Q_{U_t|X_t}(u_t|x_t)}{R_{U_t|X_t}(u_t|x_t)}.\end{aligned} \tag{7.8}$$

Notice that this cost is minimized when $Q_{U_t|X_t}=R_{U_t|X_t}$.

7.1.3 Discrete-time discrete-state KL control problem

We are now ready to state the KL control problem over the discrete-time and discrete-state setting. Introducing a positive weighting factor $\alpha>0$, it is formulated as an optimization problem

$$\min_{\{Q_{U_t|X_t}\}_{t=0}^{T-1}} \mathbb{E}^Q\sum_{t=0}^{T-1}\Big\{C_t(X_t,U_t)+\alpha D(Q_{U_t|X_t}(\cdot|X_t)\|R_{U_t|X_t}(\cdot|X_t))\Big\}+\mathbb{E}^Q C_T^{\text{exit}}(X_T). \tag{7.9}$$

[1]More generally, it is possible to consider the state-dependent cost of the form $\mathbb{E}C_t(X_t,U_t,X_{t+1})$. In this note, we restrict our attention to the state-dependent cost of the form (7.6) for simplicity.

Here, $C_T^{\text{exit}}(\cdot) : \mathcal{X} \to \mathbb{R}$ is a given terminal cost function. The problem (7.9) can be written more explicitly as[2]

$$\min_{\{Q(u_t|x_t)\}_{t=0}^{T-1}} \sum_{t=0}^{T-1} \sum_{x_t,u_t} Q(x_t)Q(u_t|x_t)\left\{C_t(x_t,u_t) + \alpha \log \frac{Q(u_t|x_t)}{R(u_t|x_t)}\right\} + \sum_{x_T} Q_{X_T}(x_T) C_T^{\text{exit}}(x_T). \tag{7.10}$$

where Q_{X_t} is sequentially computed by the recursive formula (7.5). Introducing a short-hand notation

$$C_{k:T}(x_{k:T}, u_{k:T-1}) = \sum_{t=k}^{T-1} C_t(x_t,u_t) + C_T^{\text{exit}}(x_T), \tag{7.11}$$

(7.10) can also be written as

$$\min_{\{Q_{U_t|X_t}\}_{t=0}^{T-1}} \sum_{x_{0:T}} \sum_{u_{0:T-1}} Q_{X_{0:T},U_{0:T-1}}(x_{0:T}, u_{0:T-1}) \times \left\{C_{0:T}(x_{0:T}, u_{0:T-1}) + \alpha \log \frac{Q_{X_{0:T},U_{0:T-1}}(x_{0:T}, u_{0:T-1})}{R_{X_{0:T},U_{0:T-1}}(x_{0:T}, u_{0:T-1})}\right\} \tag{7.12}$$

or more compactly as

$$\min_{\{Q_{U_t|X_t}\}_{t=0}^{T-1}} \mathbb{E}^Q \left[C_{0:T}(X_{0:T}, U_{0:T-1})\right] + \alpha D(Q_{X_{0:T},U_{0:T-1}} \| R_{X_{0:T},U_{0:T-1}}). \tag{7.13}$$

We can view (7.9) as the "stage-wise representation" of the KL control problem as it emphasizes the stage-additive structure of the cost functional. On the other hand, (7.13) can be viewed as the "trajectory-level representation" of the KL control problem, as it can be interpreted as a problem of optimizing the closed-loop trajectory distribution $Q_{X_{0:T},U_{0:T-1}}$ with respect to the nominal trajectory distribution $R_{X_{0:T},U_{0:T-1}}$.

Exercise 7.1 *Verify that* (7.10) *can be written as* (7.12).

7.1.4 Solution via backward dynamic programming

Leveraging the stage-additive structure of equation (7.9), we begin by applying the dynamic programming principle to the KL control problem to gain insight into its optimal policy. While dynamic programming is a broadly applicable solution strategy for optimal control problems with

[2] We adopt the following simplified notation:

$$\begin{aligned}
&Q_{X_t}(x_t) \to Q(x_t)\\
&Q_{U_t|X_t}(u_t|x_t) \to Q(u_t|x_t)\\
&R_{U_t|X_t}(u_t|x_t) \to R(u_t|x_t)\\
&P_{X_{t+1}|X_t,U_t}(x_{t+1}|x_t,u_t) \to P(x_{t+1}|x_t,u_t).
\end{aligned}$$

fully observable state spaces [2], its computational cost can be prohibitive, especially as the size of the state space increases – a phenomenon known as the curse of dimensionality. However, KL control represents a special class of problems where dynamic programming can be executed exactly and with remarkably low computational cost. Specifically, solving a discrete-time, discrete-state KL control problem requires only a sequence of linear operations (namely, additions and matrix multiplications) performed backward in time. This efficiency is why KL control problems are often referred to as *linearly solvable MDPs* in the literature [1].

For each $t \in \mathcal{T}$ and $x_t \in \mathcal{X}_t$, define the value function by

$$V_t(x_t) \triangleq \min_{\{Q(u_k|x_k)\}_{k=t}^{T-1}} \sum_{k=t}^{T-1} \sum_{x_k, u_k} Q(x_k)Q(u_k|x_k) \left\{ C_k(x_k, u_k) + \alpha \log \frac{Q(u_k|x_k)}{R(u_k|x_k)} \right\} + \sum_{x_T} Q_{X_T}(x_T) C_T^{\text{exit}}(x_T) \tag{7.14}$$

where $\{Q_{X_k}\}_{k=t}^{T}$ is computed using (7.5) with the policy $\{Q_{U_k|X_k}\}_{k=t,\dots,T-1}$ with the initial condition

$$Q_{X_t}(x) = \mathbb{1}_{x_t}(x) = \begin{cases} 1 & \text{if } x = x_t \\ 0 & \text{otherwise.} \end{cases} \tag{7.15}$$

The value function satisfies the following Bellman's equation:

$$V_t(x_t) = \min_{Q(u_t|x_t)} \sum_{u_t} Q(u_t|x_t) \Bigg\{ C_t(x_t, u_t) + \alpha \log \frac{Q(u_t|x_t)}{R(u_t|x_t)} + \sum_{x_{t+1}} P(x_{t+1}|x_t, u_t) V_{t+1}(x_{t+1}) \Bigg\} \tag{7.16}$$

with the terminal condition $V_T(x_T) = C_T^{\text{exit}}(x_T)$.

Exercise 7.2 *Verify equation* (7.16).

Equation (7.16) implies that the value function V can be computed backward in time by solving the minimization problem on the right hand side of (7.16) for each $x_t \in \mathcal{X}$. For each $x_t \in \mathcal{X}$, the corresponding optimal state feedback control law $Q_{U_t|X_t}$ is obtained as the minimizer of the right-hand side of (7.16).

What is noteworthy about the KL control problem is that its structure allows us to simplify the right-hand side of (7.16). Let's show this by backward induction. First, note that an explicit expression for $V_t(\cdot)$ for $t = T$ is available as $V_T(\cdot) = C_T^{\text{exit}}(\cdot)$. Next, let us show that if an analytical expression for $V_{k+1}(\cdot)$ is available, then an analytical expression for $V_k(\cdot)$ is also available. To see this, introduce a function

$$\rho_k(x_k, u_k) := C_k(x_k, u_k) + \sum_{x_{t+1}} P(x_{t+1}|x_t, u_t) V_{k+1}(x_{k+1}). \tag{7.17}$$

Then, Bellman's equation (7.16) can be written as

$$V_k(x_k) = \min_{Q(u_k|x_k)} \sum_{u_k} Q(u_k|x_k) \left\{ \rho_k(x_k, u_k) + \alpha \log \frac{Q(u_k|x_k)}{R(u_k|x_k)} \right\}. \tag{7.18}$$

From Exercise 1 in Chapter 6, we know that the unique minimizer to the right hand side of (7.18) is given by

$$Q^*(u_k|x_k) = \frac{R(u_k|x_k)\exp(-\rho_k(x_k,u_k)/\alpha)}{\sum_{u'_k} R(u'_k|x_k)\exp(-\rho_k(x_k,u'_k)/\alpha)} \tag{7.19}$$

and the optimal value is

$$V_k(x_k) = -\alpha \log \sum_{u_k} R_{U_k|X_k}(u_k|x_k)\exp\left(-\frac{\rho_k(x_k,u_k)}{\alpha}\right). \tag{7.20}$$

Since C_k and $R_{U_k|X_k}$ are known quantities, equations (7.17) and (7.20) above provide a recursive method to compute the value functions $V_k(\cdot)$ backward in time. The algorithm is summarized in Algorithm 11.

Algorithm 11: Optimal KL control synthesis by backward dynamic programming

Define $V_T \in \mathbb{R}^{|\mathcal{X}_T|}$ by $V_T(x_T) = C_T^{\text{exit}}(x_T)$ for each $x_T \in \{1,2,...,|\mathcal{X}_T|\}$;
for $k = T-1, T-2, \cdots, 2, 1$ **do**
  Compute $\rho_k(x_k,u_k)$ using (7.17) ;
  Compute the optimal policy $Q^*(u_k|x_k)$ using (7.19) ;
  Compute the value function $V_k(x_k)$ using (7.20) ;

7.1.5 Linearly solvable KL control

Now we focus on the special case in which the state transition law is deterministic, i.e., the next state x_{t+1} is a deterministic function of x_t and u_t as $x_{t+1} = F(x_t, u_t)$. This special case is noteworthy as it leads to a further simplification of the backward dynamic programming algorithm summarized in Algorithm 11. The resulting KL control problem is known as linearly solvable KL control since only a linear operation (i.e., a matrix multiplication) is needed in each state of the backward dynamic programming. This special case is also important as it allows a completely different class of solution algorithms based on forward-in-time Monte-Carlo simulation, which we will discuss in the next section (Section 7.1.6).

Deterministic transition

Specifically, we assume:

Assumption 7.1 *(Deterministic transition) The state transition law is given by*

$$P_{X_{t+1}|X_t,U_t}(x_{t+1}|x_t,u_t) = \mathbb{1}_{\{x_{t+1}=F_t(x_t,u_t)\}}(x_{t+1}) = \begin{cases} 1 & \textit{if } x_{t+1} = F_t(x_t,u_t) \\ 0 & \textit{otherwise} \end{cases} \tag{7.21}$$

where $F_t : \mathcal{X}_t \times \mathcal{U}_t \to \mathcal{X}_{t+1}$ is a given mapping.[3]

[3] In the literature (e.g., [3]), it is common to adopt a simplified state transition model in which $\mathcal{U}_t = \mathcal{X}_{t+1}$ and

$$P(x_{t+1}|x_t,u_t) = \begin{cases} 1 & \text{if } x_{t+1} = u_t \\ 0 & \text{otherwise.} \end{cases}$$

To see how Assumption 7.1 simplifies Algorithm 11, notice that Assumption 7.1 allows us to write (7.17) as

$$\rho_k(x_k, u_k) := C_k(x_k, u_k) + V_{k+1}(F_k(x_k, u_k)). \tag{7.22}$$

Let us now introduce the following logarithmic transformation to express the value function V_k in terms of a new variable Z_k:

$$V_k(x_k) = -\alpha \log Z_k(x_k). \tag{7.23}$$

In what follows, we call Z_k the *exponentiated value function* or the *desirability function*. Using Z_k, (7.20) can be written as

$$Z_k(x_k) = \sum_{u_k} R(u_k|x_k) \exp\left(-\frac{\rho_k(x_k, u_k)}{\alpha}\right). \tag{7.24}$$

Substituting (7.22) into (7.24), we obtain

$$Z_k(x_k) = \sum_{u_k} R(u_k|x_k) \exp\left(-\frac{C_k(x_k, u_k)}{\alpha}\right) Z_{k+1}(F_k(x_k, u_k)). \tag{7.25}$$

Notably, (7.25) indicates a linear relationship between Z_k and Z_{k+1}. More explicitly, if we represent the function $Z_k(\cdot) : \mathcal{X}_k \to \mathbb{R}$ as an $|\mathcal{X}_k|$-dimensional vector

$$Z_k = \begin{bmatrix} Z_k(1) & \cdots & Z_k(|\mathcal{X}_k|) \end{bmatrix}^\top \in \mathbb{R}^{|\mathcal{X}_k|},$$

equation (7.25) can be written as

$$Z_k = M_k Z_{k+1} \tag{7.26}$$

where $M_k \in \mathbb{R}^{|\mathcal{X}_k| \times |\mathcal{X}_{k+1}|}$ is a matrix whose (x, x')-th entry is defined by

$$M_k(x, x') = \sum_u \mathbb{1}_{\{x'=F_k(x,u)\}} R(u|x) \exp\left(-\frac{C_k(x, u)}{\alpha}\right). \tag{7.27}$$

Thus, we have obtained an important conclusion that the discrete-time discrete-state KL control problem can be solved simply by performing sequential matrix multiplications (7.26) backward in time, starting from the terminal condition:

$$Z_T(x_T) = \exp\left(-\frac{C_T^{\text{exit}}(x_T)}{\alpha}\right). \tag{7.28}$$

Once the functions $Z_t(\cdot)$ are computed for all $t \in \mathcal{T}$, the optimal policy can be reconstructed as

$$Q^*(u_k|x_k) = R(u_k|x_k) \exp\left(-\frac{C_k(x_k, u_k)}{\alpha}\right) \frac{Z_{k+1}(F_k(x_k, u_k))}{Z_k(x_k)}. \tag{7.29}$$

The simplified backward dynamic programming is summarized in Algorithm 12.

This assumption means that the policy designer has full control of the state transition probability and is a special case of Assumption 7.1. Under this assumption, the state transition equation (7.5) can be simplified to

$$Q(x_{t+1}) = \sum_{x_t \in \mathcal{X}_t} Q(x_{t+1}|x_t) Q(x_t)$$

and $\{Q(x_{t+1}|x_t)\}_{t=0}^{T-1}$ becomes the policy to be designed. Although such an assumption is effective to simplify the problem formulation, it is restrictive in many applications. Thus, in our problem formulation, we assume that $\mathcal{U}_t$ and $\mathcal{X}_{t+1}$ are different sets in general.

Algorithm 12: Linearly solvable KL control: backward dynamic programming approach

Define $Z_T \in \mathbb{R}^{|\mathcal{X}_T|}$ by $Z_T(x_T) = \exp\left(-\frac{C_T^{\text{exit}}(x_T)}{\alpha}\right)$ for each $x_T \in \{1, 2, ..., |\mathcal{X}_T|\}$;
for $k = T-1, T-2, \cdots, 2, 1$ **do**
  Compute $Z_k = M_k Z_{k+1}$ using the matrix M_k is defined by equation (7.27) ;
  Compute the optimal policy $Q^*_{U_k|X_k}$ by (7.29) ;

Note that the determinism of the transition model (Assumption 7.1) is critical for the KL control problem to be linearly solvable. To see this, let us analyze how the removal of Assumption 7.1 alters the linear relationship (7.26).

Necessity of determinism

Recall that under the general transition model $P(x_{k+1}|x_k, u_k)$, $\rho_k(x_k, u_k)$ is expressed as (7.17). Substituting (7.17) into (7.24), we obtain

$$\begin{aligned} Z_k(x_k) &= \sum_{u_k} R(u_k|x_k) \exp\left(-\frac{C_k(x_k, u_k)}{\alpha}\right) \exp\left(-\frac{1}{\alpha}\sum_{x_{k+1}} P(x_{k+1}|x_k, u_k) V_{k+1}(x_{k+1})\right) \\ &\leq \sum_{u_k} R(u_k|x_k) \exp\left(-\frac{C_k(x_k, u_k)}{\alpha}\right) \sum_{x_{k+1}} P(x_{k+1}|x_k, u_k) \exp\left(-\frac{V_{k+1}(x_{k+1})}{\alpha}\right) \\ &= \sum_{u_k} R(u_k|x_k) \exp\left(-\frac{C_k(x_k, u_k)}{\alpha}\right) \sum_{x_{k+1}} P(x_{k+1}|x_k, u_k) Z_{k+1}(x_{k+1}). \end{aligned}$$

Note that we have applied Jensen's inequality in the second line, which holds with equality if and only if $P(x_{k+1}|x_k, u_k)$ corresponds to a deterministic transition. Introducing a matrix $M_k \in \mathbb{R}^{|\mathcal{X}_k| \times |\mathcal{X}_{k+1}|}$ is a matrix whose (x, x')-th entry is

$$M_k(x, x') = \sum_u P(x'|x, u) R(u|x) \exp\left(-\frac{C_k(x, u)}{\alpha}\right), \tag{7.30}$$

the inequality above can be written as

$$Z_k \leq M_k Z_{k+1}. \tag{7.31}$$

The result (7.31) indicates that there exists a linear *inequality* relationship between the desirability functions Z_k and Z_{k+1} in general, with the equality holding if and only if the transition model is deterministic (Assumption 7.1).

Exercise 7.3 *(KL control problems that are not linearly solvable) Using the matrix M_k defined by* (7.30)*, suppose we perform the backward linear recursion $\tilde{Z}_k = M_k \tilde{Z}_{k+1}$ starting from the terminal condition $\tilde{Z}_T(x_T) = \exp\left(-\frac{C_T^{\text{exit}}(x_T)}{\alpha}\right)$. Show that $Z_t \leq \tilde{Z}_t$ holds entry-wise for each $t \in \mathcal{T}$.*

7.1.6 Solution via forward Monte-Carlo simulation

In the previous section, we derived a linear relationship between Z_k and Z_{k+1} specified by

$$Z_k(x_k) = \sum_{u_k} R_{U_k|X_k}(u_k|x_k) \exp\left(-\frac{C_k(x_k, u_k)}{\alpha}\right) Z_{k+1}(x_{k+1}). \tag{7.32}$$

where $x_{k+1} = F_k(x_k, u_k)$. The structure of (7.32) naturally indicates that the desirability function Z_k can be computed by a backward recursion. In this section, we will see that (7.32) implies that the desirability function Z_k can also be computed by forward-in-time Monte-Carlo simulations. This observation is the essence of the path integral control algorithm we will see in Chapter VIII.

Free-energy representation of the value function

To develop an alternative method to compute the right-hand side of (7.32), note that repetitive substitutions yield

$$\begin{aligned} Z_k(x_k) =& \sum_{u_k} R(u_k|x_k) \exp\left(-\frac{C_k(x_k, u_k)}{\alpha}\right) \times \cdots \\ & \times \sum_{u_{T-1}} R(u_{T-1}|x_{T-1}) \exp\left(-\frac{C_{T-1}(x_{T-1}, u_{T-1})}{\alpha}\right) \\ & \times \exp\left(-\frac{C_T^{\text{exit}}(x_T)}{\alpha}\right) \end{aligned} \tag{7.33}$$

where $x_{t+1} = F_t(x_t, u_t)$ for $t = k, \cdots, T-1$. Let us introduce a short-hand notation for the joint distribution:[4]

$$R(x_{k+1:T}, u_{k:T-1}|x_k) := \prod_{t=k}^{T-1} R(u_t|x_t) \mathbb{1}_{\{x_{t+1}=F_t(x_t,u_t)\}}. \tag{7.34}$$

Using the notation $C_{k:T}(x_{k:T}, u_{k:T-1})$ for the path cost defined in (7.11), (7.33) can be expressed more compactly as

$$\begin{aligned} Z_k(x_k) &= \sum_{x_{k+1:T}} \sum_{u_{k:T-1}} R(x_{k+1:T}, u_{k:T-1}|x_k) \exp\left(-\frac{C_{k:T}(x_{k:T}, u_{k:T-1})}{\alpha}\right) \\ &=: \mathbb{E}^R \exp\left(-\frac{C_{k:T}(x_{k:T}, u_{k:T-1})}{\alpha}\right). \end{aligned} \tag{7.35}$$

Therefore, from (7.23), the value function is written as

$$V_k(x_k) = -\alpha \log \left\{ \mathbb{E}^R \exp\left(-\frac{C_{k:T}(x_{k:T}, u_{k:T-1})}{\alpha}\right) \right\}. \tag{7.36}$$

One can view (7.36) as the free energy representation of the value function. Specifically, for $k = 0$, we notice that (7.36) is nothing but the result of applying the variational formula to the optimization problem (7.13).

[4] The indicator function $\mathbb{1}_{\{x_{t+1}=F_t(x_t,u_t)\}}$ ensures that the transition law $x_{t+1} = F_t(x_t, u_t)$ is imposed for $t = k, \cdots, T-1$ in the joint distribution.

Monte-Carlo algorithm

Equation (7.35) implies that the desirability function $Z_k(x_k)$ can be computed by evaluating the path probability $R(x_{k+1:T}, u_{k:T-1}|x_k)$ and the path cost $C_{k:T}$ for all possible paths starting from x_k at $t = k$. However, since there are $|\mathcal{U}_k| \times \cdots \times |\mathcal{U}_{T-1}|$ possible paths, such a naive approach does not seem scalable.

Alternatively, one can apply the Monte Carlo method to approximate the value of $Z_k(x_k)$. Suppose we generate N sample paths $x_k \overset{u_k}{\to} x_{k+1} \overset{u_{k+1}}{\to} \cdots \overset{u_{T-1}}{\to} x_T$, all starting from x_k at time step $t = k$, independently according to the reference control policy[5] $\{R(u_t|x_t)\}_{t=k}^{T-1}$. Denote the paths obtained by the i-th experiment by $(X_{k:T}(i), U_{k:T-1}(i))$. (Here, we use an uppercase since it is a random variable.) For each sample path, let us also compute the corresponding path cost $C_{k:T}(X_{k:T}(i), U_{k:T-1}(i))$. Let

$$r(x_k) \triangleq \frac{1}{N}\sum_{i=1}^{N} \exp\left(-\frac{C_{k:T}(X_{k:T}(i), U_{k:T-1}(i))}{\alpha}\right) \tag{7.37}$$

be the empirical desirability of the state x_k. Since we have (7.35), by the strong law of large numbers, we obtain

$$r(x_k) \overset{a.s.}{\to} Z_k(x_k) \text{ as } N \to \infty. \tag{7.38}$$

Once $Z_k(x_k)$ and $Z_{k+1}(x_{k+1})$ are obtained, the optimal policy $Q^*(u_k|x_k)$ can be computed using the formula (7.29). Note that $Q^*(u_k|x_k)$ is a randomized policy in general – that is, for each x_k, the action u_k must be sampled from the distribution $Q^*(u_k|x_k)$. How can we achieve this algorithmically? Can we use Monte Carlo simulations again to complete this task?

The rest of this section introduces a Monte Carlo-based scheme to sample u_k from $Q^*(u_k|x_k)$. Among the N sample paths, let $\mathcal{I}(u_k)$ be the collection of the indices of the paths in which the action $u_k \in \mathcal{U}_k$ is taken at time step $t = k$. Clearly, $\sum_{u_k \in \mathcal{U}_k} |\mathcal{I}(u_k)| = N$ holds, while for each $u_k \in \mathcal{U}_k$, $|\mathcal{I}(u_k)|$ is a random variable. Moreover, we have

$$\frac{|\mathcal{I}(u_k)|}{N} \overset{a.s.}{\to} R_{U_k|X_k}(u_k|x_k) \text{ as } N \to \infty \tag{7.39}$$

by the strong law of large numbers. For each $u_k \in \mathcal{U}_k$, let

$$r(x_k, u_k) \triangleq \frac{1}{N}\sum_{i\in\mathcal{I}(u_k)} \exp\left(-\frac{C_{k:T}(X_{k:T}(i), U_{k:T-1}(i))}{\alpha}\right) \tag{7.40}$$

be the empirical reward of taking u_k. From (7.37) and (7.40), clearly we have

$$r(x_k) = \sum_{u_k \in \mathcal{U}_k} r(x_k, u_k). \tag{7.41}$$

The following convergence result holds for $r(x_k, u_k)$:

[5]Since $\{R(u_t|x_t)\}_{t=k}^{T-1}$ is known, it is easy to generate a collection of such sample paths using a Monte-Carlo simulation.

Exercise 7.4 *Show that*

$$r(x_k, u_k) \overset{a.s.}{\to} R_{U_k|X_k}(u_k|x_k) \exp\left(-\frac{C_k(x_k, u_k)}{\alpha}\right) Z_{k+1}(F_k(x_k, u_k)) \tag{7.42}$$

as $N \to \infty$.

Now, suppose that x_k is the realization of the state at time step $t = k$. To implement a KL controller at this state, we generate N sample paths starting from x_k under the reference policy $\{R_{U_t|X_t}\}_{t=k}^{T-1}$. We then compute the empirical reward $r(x_k, u_k)$ for each $u_k \in \mathcal{U}_k$ and evaluate the ratio $\frac{r(x_k,u_k)}{\sum_{u_k} r(x_k,u_k)}$. Combining the results (7.38) and (7.42), it is easy to see that

$$\frac{r(x_k, u_k)}{\sum_{u_k} r(x_k, u_k)} = \frac{r(x_k, u_k)}{r(x_k)} \tag{7.43a}$$

$$\overset{a.s.}{\to} R_{U_k|X_k}(u_k|x_k) \exp\left(-\frac{C_k(x_k, u_k)}{\alpha}\right) \frac{Z_{k+1}(F_k(x_k, u_k))}{Z_k(x_k)} \tag{7.43b}$$

$$= Q^*_{U_k|X_k}(u_k|x_k) \tag{7.43c}$$

as $N \to \infty$. The convergence result (7.43) above suggests how to approximate $Q^*_{U_k|X_k}(u_k|x_k)$ using Monte-Carlo simulations. The control synthesis based on Monte-Carlo simulation is summarized in Algorithm 13.

Algorithm 13: Optimal KL control synthesis by forward Monte-Carlo simulations

1 Given: Initial state x_0;

2 **for** $k = 0, 1, ..., T-1$ **do**

3 Generate N sample paths $X_{k:T}(i), U_{k:T-1}(i)$ for $i = 1, 2, ..., N$ starting from $x_k \in \mathcal{X}$ randomly according to the reference state transition policy $\{R_{U_k|X_k}\}_{t=k,...,T-1}$;

4 For each $u_k \in \mathcal{U}_k$, find the subset of the sample paths in which the action u_k is taken at time step $t = k$. Let $\mathcal{I}(u_k)$ be the set of index of such paths;

5 For each $u_k \in \mathcal{U}_k$, compute the empirical reward $r(x_k, u_k)$ by $r(x_k, u_k) = \frac{1}{N} \sum_{i \in \mathcal{I}(u_k)} \exp\left(-\frac{C_{k:T}(X_{k:T}(i), U_{k:T-1}(i))}{\alpha}\right)$;

6 For each $u_k \in \mathcal{U}_k$, compute the approximation $Q^{\text{approx}}_{U_k|X_k}(u_k|x_k)$ of the optimal policy $Q^*_{U_k|X_k}(u_k|x_k)$ by $Q^{\text{approx}}_{U_k|X_k}(u_k|x_k) = \frac{r(x_k,u_k)}{\sum_{u'_k \in \mathcal{U}_k} r(x_k,u'_k)}$;

7 Select a control action u_k according to the policy $Q^{\text{approx}}_{U_k|X_k}(u_k|x_k)$;

Exercise 7.5 *How can we execute Line 7 of Algorithm 13? Hint: Consider the "inverse transform method" summarized in Algorithm 10.*

Exercise 7.6 *What is the potential difficulty of applying Monte Carlo algorithms to solve KL control problems without the deterministic transition property (Assumption 7.1)*

In Algorithm 13, we construct the conditional probability distribution $Q(u_k|x_k)$ explicitly in Line 6, from which an action u_k is sampled in Line 7. Note that such a construction is possible thanks to the underlying assumption of this section that $\mathcal{U}_t$ is a finite space, and that $Q(u_k|x_k)$ is an $|\mathcal{U}_k|$-dimensional stochastic vector. However, if $\mathcal{X}_t$ and $\mathcal{U}_t$ are continuous spaces, an explicit construction of the conditional probability distribution $Q(u_k|x_k)$ is not possible. Alternative strategies to sample from $Q(u_k|x_k)$ will be necessary. This will be considered in the next section.

7.2 Problem formulation: Discrete-time, continuous-state case

As in the previous chapter, consider a stochastic optimal control problem over the discrete time index $\mathcal{T} = \{0, 1, ..., T\}$. The state random process $\{X_t\}_{t\in\mathcal{T}}$ is a sequence of $(\mathcal{X}_t, \mathcal{B}(\mathcal{X}_t))$-valued random variables, whereas the control random process $\{U_t\}_{t=0}^{T-1}$ is a sequence of $(\mathcal{U}_t, \mathcal{B}(\mathcal{U}_t))$-valued random variables. We assume that both $\mathcal{X}_t$ and $\mathcal{U}_t$ are subsets of $\mathbb{R}^n$, and that $\mathcal{B}(\mathcal{X}_t)$ and $\mathcal{B}(\mathcal{U}_t)$ denote the Borel σ-algebra. Let us first introduce the fixed end-time formulation, followed by the variable end-time formulation.

For each $t \in \{0, 1, ..., T-1\}$, let the stochastic kernel $Q_{U_t|X_t}$ on $\mathcal{U}_t$ given $\mathcal{X}_t$ represent the control policy to be designed. If the initial state distribution P_{X_0} is given and the control policy $\{Q_{U_t|X_t}\}_{t=0}^{T-1}$ is fixed, the probability measure $Q_{X_{0:T},U_{0:T-1}}$ on the space of trajectories

$$(\mathcal{X}_0 \times \cdots \times \mathcal{X}_T \times \mathcal{U}_0 \times \cdots \times \mathcal{U}_{T-1}, \mathcal{B}(\mathcal{X}_0 \times \cdots \times \mathcal{X}_T \times \mathcal{U}_0 \times \cdots \times \mathcal{U}_{T-1}))$$

is defined by

$$\begin{aligned} &Q_{X_{0:T},U_{0:T-1}}(B_{X_0} \times \cdots \times B_{X_T} \times B_{U_0} \times \cdots \times B_{U_T}) \\ &= \int_{B_{X_0}} \int_{B_{U_0}} \cdots \int_{B_{U_{T-1}}} \int_{B_{X_T}} \prod_{t=0}^{T-1} P(dx_{t+1}|x_t, u_t) Q(du_t|x_t) P(dx_0) \end{aligned} \tag{7.44}$$

for each Cartesian product of measurable sets $B_{X_0}, \cdots, B_{X_T}, B_{U_0}, \cdots, B_{U_T}$. The instantaneous state distribution $Q_{X_t}(B_{X_t})$ can be computed by marginalizing the joint distribution (7.44). Alternatively, it can also be computed recursively by

$$Q_{X_{t+1}}(B_{X_{t+1}}) = \int_{x_t\in\mathcal{X}_t} \int_{u_t\in\mathcal{U}_t} P(B_{X_{t+1}}|x_t, u_t) Q(du_t|x_t) Q(dx_t) \tag{7.45}$$

for each $B_{X_{t+1}} \in \mathcal{B}(\mathcal{X}_{t+1})$. Suppose Borel measurable functions $C_t(\cdot,\cdot) : \mathcal{X}_t \times \mathcal{U}_t \to \mathbb{R}$ for $t = 0, 1, ..., T-1$ and $C_T^{\text{exit}}(\cdot) : \mathcal{X}_T \to \mathbb{R}$ represent the stage-wise and the terminal cost functions, respectively. Suppose also that the reference state transition policy is given as a sequence of stochastic kernels $R_{U_t|X_t}$. Introducing a positive weighting factor α, the discrete-time continuous-state KL control problem is formulated as

$$\min_{\{Q_{U_t|X_t}\}_{t=1}^{T-1}} \mathbb{E} \sum_{t=0}^{T-1} \Big\{ C_t(X_t, U_t) + \alpha D(Q_{U_t|X_t}(\cdot|X_t) \| R_{U_t|X_t}(\cdot|X_t)) \Big\} + \mathbb{E} C_T(X_T) \tag{7.46}$$

where the minimization is over the space of stochastic kernels $Q_{U_t|X_t}$ on $\mathcal{U}_t$ given $\mathcal{X}_t$ for $t = 0, 1, ..., T-1$. More precisely, (7.46) means

$$\min_{\{Q_{U_t|X_t}\}_{t=1}^{T-1}} \sum_{t=0}^{T-1} \int_{x_t \in \mathcal{X}_t} \int_{u_t \in \mathcal{U}_t} \left\{ C_t(x_t, u_t) + \alpha \log \left(\frac{dQ_{U_t|X_t}}{dR_{U_t|X_t}}(x_t, u_t)(x_t, u_t) \right) \right\} Q(du_t|x_t) Q(dx_t) + \int_{x_T \in \mathcal{X}_T^{\text{exit}}} C_T^{\text{exit}}(x_T)(dx_T) \tag{7.47}$$

where the measure Q_{X_t} is computed by (7.45). In (7.47), $\frac{dQ_{U_t|X_t}}{dR_{U_t|X_t}}(x_t, u_t)$ is the Radon-Nikodym derivative. That is, for each fixed x_t, it is a function satisfying

$$\int_{B_{U_t}} g(u_t) Q(du_t|x_t) = \int_{B_{U_t}} g(u_t) \frac{dQ_{U_t|X_t}}{dR_{U_t|X_t}}(x_t, u_t) R(du_t|x_t) \tag{7.48}$$

for each $B_{U_t} \in \mathcal{B}(\mathcal{U}_t)$ and a $\mathcal{B}(\mathcal{U}_t)$-measurable function $g(u_t)$.

In Section 7.1.3, we have seen that a discrete-time discrete-state KL control problem admits both a stage-wise representation (7.9) and a trajectory-level representation (7.13). In a similar vein, a discrete-time continuous-state KL control problem admits both a stage-wise representation (7.46) and a trajectory-level representation:

$$\min_{\{Q_{U_t|X_t}\}_{t=0}^{T-1}} \mathbb{E} \left\{ C_{0:T}(X_{0:T}, U_{0:T-1}) + \alpha D(Q_{X_{0:T}, U_{0:T-1}} \| R_{X_{0:T}, U_{0:T-1}}) \right\} \tag{7.49}$$

where

$$C_{0:T}(x_{0:T}, u_{0:T-1}) = \sum_{t=0}^{T-1} C_t(x_t, u_t) + C_T^{\text{exit}}(x_T). \tag{7.50}$$

The next exercise confirms the equivalence between (7.46) and (7.49).

Exercise 7.7 *Notice that* (7.49) *can be written more explicitly as*

$$\min_{\{Q_{U_t|X_t}\}_{t=0}^{T-1}} \int_{\mathcal{X}_{0:T}} \int_{\mathcal{U}_{0:T-1}} \left\{ C_{0:T}(x_{0:T}, u_{0:T-1}) + \alpha \log \frac{dQ_{X_{0:T}, U_{0:T-1}}}{dR_{X_{0:T}, U_{0:T-1}}}(x_{0:T}, u_{0:T-1}) \right\} \times Q_{X_{0:T}, U_{0:T-1}}(dx_{0:T} \times du_{0:T-1}) \tag{7.51}$$

Prove that

$$\frac{dQ_{X_{0:T}, U_{0:T-1}}}{dR_{X_{0:T}, U_{0:T-1}}} = \prod_{t=0}^{T-1} \frac{dQ_{U_t|X_t}}{dR_{U_t|X_t}} \tag{7.52}$$

holds. Using (7.52)*, show that* (7.47) *and* (7.51) *are equivalent.*

7.2.1 Solution via backward dynamic programming

As in the previous chapter, we are particularly interested in the case where the state transition law $P(x_{t+1}|x_t, u_t)$ is governed by a deterministic mapping $x_{t+1} = F(x_t, u_t)$. Specifically, let us make the following assumption:

Assumption 7.2 *(Deterministic transition) The state transition law is given by*

$$P(x_{t+1}|x_t, u_t) = \delta_{F_t(x_t,u_t)}(x_{t+1}) \tag{7.53}$$

where δ denotes the Dirac measure and $F_t : \mathcal{X}_t \times \mathcal{U}_t \to \mathcal{X}_{t+1}$ is a given measurable mapping.

For each $t \in \mathcal{T}$ and $x_t \in \mathcal{X}_t$, define the value function $V : \mathcal{T} \times \mathcal{X}_t \to \mathbb{R}$ by

$$V_t(x_t) \triangleq \inf_{\{Q(u_k|x_k)\}_{k=t}^{T-1}} \sum_{k=t}^{T-1} \Bigg[\int_{x_k \in \mathcal{X}_k} \int_{u_k \in \mathcal{U}_k} \left\{ C_k(x_k, u_k) + \alpha \log \left(\frac{dQ_{U_k|X_k}}{dR_{U_k|X_k}}(u_k|x_k) \right) \right\} Q(du_k|x_k) Q(dx_k)$$
$$+ \int_{x_T \in \mathcal{X}_T} C_T^{\text{exit}}(x_T) Q(dx_T) \Bigg]. \tag{7.54}$$

Notice that we used "inf" instead of "$\min$" in (7.54) as we have not yet proved that the infimum is attained. In the sequel, we will show that infimum is indeed attained in general and thus "inf" can be replaced with "$\min$". It follows from the definition (7.54) that the value function satisfies $V_T(x_T) = C_T^{\text{exit}}(x_T)$ and

$$V_t(x_t) = \inf_{Q_{U_t|X_t}} \int_{u_t \in \mathcal{U}_t} \left\{ \rho_t(x_t, u_t) + \alpha \log \left(\frac{dQ_{U_t|X_t}}{dR_{U_t|X_t}}(u_t|x_t) \right) \right\} Q_{U_t|X_t}(du_t|x_t). \tag{7.55}$$

In (7.55), we have introduced the function

$$\rho_t(x_t, u_t) \triangleq C_t(x_t, u_t) + V_{t+1}(F_t(x_t, u_t)). \tag{7.56}$$

Now, we need to study the solution to the minimization problem appearing in the right hand side of (7.55).

Applying the result of Exercise 3 in Chapter VI, one can see that the infimum of the right hand side of (7.55) is attained by the stochastic kernel $Q^*(u_t|x_t)$ characterized by

$$Q^*(B_{U_t}|x_t) = \frac{\int_{B_{U_t}} \exp(-\rho_t(x_t, u_t)/\alpha) R(du_t|x_t)}{\int_{\mathcal{U}_t} \exp(-\rho_t(x_t, u_t)/\alpha)(du_t|x_t)} \tag{7.57}$$

and that the infimum is explicitly computed as

$$V_t(x_t) = -\alpha \log \left\{ \int_{\mathcal{U}_t} \exp\left(-\frac{\rho_t(x_t, u_t)}{\alpha} \right) R(du_t|x_t) \right\}.$$

Substituting the definition (7.56), we obtain a recursive expression of the value function $V_t(\cdot)$ as follows:

$$V_t(x_t) = -\alpha \log \left\{ \int_{\mathcal{U}_t} \exp\left(-\frac{C_t(x_t, u_t)}{\alpha} \right) \exp\left(-\frac{V_{t+1}(F_t(x_t, u_t))}{\alpha} \right) R(du_t|x_t) \right\} \tag{7.58}$$

for $t = 1, 2, \ldots, T-1$ and $V_T(x_T) = C_T^{\text{exit}}(x_T)$. As in discrete-time discrete-space cases, by introducing the change of variables

$$V_t(x_t) = -\alpha \log Z_t(x_t) \tag{7.59}$$

it is possible to linearize the recursion above as follows:

$$Z_t(x_t) = \int_{\mathcal{U}_t} \exp\left(-\frac{C_t(x_t, u_t)}{\alpha}\right) Z_{t+1}(F_t(x_t, u_t)) R(du_t|x_t) \tag{7.60}$$

for $t = 1, 2, \ldots, T-1$ and $Z_T(x_T) = \exp\left(-\frac{C_T^{\text{exit}}(x_T)}{\alpha}\right)$.

Once the exponentiated value functions $\{Z_t\}_{t=0}^T$ are obtained, the optimal control policy (7.57) can be expressed in terms of $\{Z_t\}_{t=0}^T$ as

$$\begin{aligned} Q^*_{U_t|X_t}(B_{U_t}|x_t) &= \frac{1}{Z_t(x_t)} \int_{B_{U_t}} \exp\left(-\frac{C_t(x_t, u_t)}{\alpha}\right) \exp\left(-\frac{V_{t+1}(F_t(x_t, u_t))}{\alpha}\right) R(du_t|x_t) \\ &= \frac{1}{Z_t(x_t)} \int_{B_{U_t}} \exp\left(-\frac{C_t(x_t, u_t)}{\alpha}\right) Z_{t+1}(F_t(x_t, u_t)) R(du_t|x_t). \end{aligned} \tag{7.61}$$

However, to perform the backward recursion (7.60), notice that the function $Z_t(x_t)$ must be evaluated everywhere in the continuous domain $\mathcal{X}_t$. Thus, despite being linear, the recursive expression (7.60) is not as computationally attractive as the discrete state counterpart we saw earlier. Therefore, instead of computing the recursion (7.60) directly, we will consider an approach based on Monte-Carlo simulation in the next section.

7.2.2 Solution via forward Monte-Carlo simulation

In Section 7.1.6, we derived a Monte-Carlo-based solution algorithm for discrete-time, discrete-state problems. We now generalize the same machinery to discrete-time, continuous-state KL control problems.

Let us first obtain a free-energy representation of the desirability function $Z_t(x_t)$ characterized by (7.59). The derivation is identical to the discussion in Section 7.1.6, except that $\sum_{\mathcal{X}}$ will be replaced by $\int_{\mathcal{X}}$. As before, we write

$$\begin{aligned} x_{k+1,T} &:= (x_{k+1}, x_{k+2}, \cdots, x_T) \\ u_{k,T-1} &:= (u_k, u_{k+1}, \cdots, x_{T-1}). \end{aligned}$$

The distribution of the paths $(x_{k+1,T}, u_{k,T-1})$ generated by the reference policy $R(u_k|x_k)$ is defined by

$$\begin{aligned} &R(B_{X_{k+1}} \times \cdots \times B_{X_T} \times B_{U_k} \times \cdots \times B_{U_{T-1}}|x_k) \\ &= \int_{B_{U_k}} \int_{B_{X_{k+1}}} \cdots \int_{B_{U_{T-1}}} \int_{B_{X_T}} \prod_{t=k}^{T-1} P(dx_{t+1}|x_t, u_t) R(du_t|x_t). \end{aligned}$$

Introduce the path cost function by

$$C_{k:T}(x_{k:T}, u_{k:T-1}) = \sum_{t=k}^{T-1} C_t(x_t, u_t) + C_T^{\text{exit}}(x_T). \tag{7.62}$$

Following the arguments similar to the discrete-state case in Section 7.1.6, we obtain

$$\begin{aligned}
&Z_k(x_k) \\
&= \int_{\mathcal{X}_{k+1:T}} \int_{\mathcal{U}_{k:T-1}} \exp\left(-\frac{C_{k:T}(x_{k:T}, u_{k:T-1})}{\alpha}\right) R(dx_{k+1:T}, du_{k:T-1}|x_k) \\
&= \mathbb{E}^R \exp\left(-\frac{C_{k:T}(x_{k:T}, u_{k:T-1})}{\alpha}\right). \qquad (7.63)
\end{aligned}$$

The last expression (7.63) looks identical to the discrete-state case (7.35). Combining (7.63) and (7.61), we obtain

$$\begin{aligned}
Q^*_{U_k|X_k}(B_{U_k}|x_k) = \frac{1}{Z_k(x_k)} &\int_{\{\mathcal{X}_{k+1:T}, \mathcal{U}_{k:T-1}|u_k \in B_{U_k}\}} \exp\left(-\frac{C_{k:T}(x_{k:T}, u_{k:T-1})}{\alpha}\right) \\
&\times R(dx_{k+1:T}, du_{k:T-1}|x_k) \qquad (7.64)
\end{aligned}$$

where $\{\mathcal{X}_{k+1:T}, \mathcal{U}_{k:T-1}|u_k \in B_{U_k}\}$ is the collection of the paths such that $u_k \in B_{U_k}$.

7.2.3 Monte-Carlo estimates of the value functions

The expression (7.63) naturally suggests the following use of Monte-Carlo simulations to estimate $Z_k(x_k)$: First, pick a sufficiently large real number N and generate a tuple of N independent realizations of the state random process $X_{k:T}$ and the control random process $U_{k:T-1}$ under the reference control policy $\{R(u_t|x_t)\}_{t=k}^{T-1}$. Denote the i-th sample path by $X_{k:T}(i)$ and $U_{k:T-1}(i)$. Second, for each realization of the sample path, compute the path cost $C_{k:T}(X_{k:T}(i), U_{k:T-1}(i))$ using (7.62). Finally, compute the empirical mean

$$r(x_k) \triangleq \frac{1}{N} \sum_{i=1}^{N} \exp\left(-\frac{C_{k:T}(X_{k:T}(i), U_{k:T-1}(i))}{\alpha}\right) \qquad (7.65)$$

By the strong law of large numbers, we expect that the empirical mean (7.65) converges to (7.63) almost surely, i.e.,

$$r(x_k) \stackrel{a.s.}{\to} Z_k(x_k) \text{ as } N \to \infty. \qquad (7.66)$$

7.2.4 Monte-Carlo estimates of the optimal control polices

To implement the optimal KL control policy numerically, one needs to be able to sample a control input u_k according to the probability distribution function $Q^*_{U_k|X_k}$ characterized by (7.64). Even though the desirability function $Z_k(x_k)$ can be approximated by Monte-Carlo simulations using (7.66), the equation (7.64) itself does not tell how to generate samples u_k from the described distribution. In discrete state cases, we introduced a method to compute the empirical reward $r(x_k, u_k)$ for each (x_k, u_k), based on which the probability mass function corresponding to the optimal policy was computed by

$$Q^*_{U_k|X_k}(u_k|x_k) \approx Q^{\text{approx}}_{U_k|X_k}(u_k|x_k) = \frac{r(x_k, u_k)}{\sum_{u'_k \in \mathcal{U}_k} r(x_k, u'_k)}. \qquad (7.67)$$

Unfortunately, (7.67) cannot be used in the current scenario since we are now considering a continuous action space $\mathcal{U}_k$. In what follows, we generalize the formula (7.67) that allows us to generate a sample u_k from the probability distribution function $Q^*_{U_k|X_k}$ characterized by (7.64).

Recall that Algorithm 10 presented in Section 6.4 allows us to generate samples from a Boltzmann distribution over a continuous space. Since (7.64) is a Boltzmann distribution, the concept of Algorithm 10 is applicable. Specifically, let us generate N independent sample paths $X_{k:t_f}(i)$, $U_{k:t_f-1}(i)$, $i = 1, 2, ..., N$ under the reference control policy $R_{U_k|X_t}$, all starting from $x_k \in \mathcal{X}_k$. For each i, let

$$r(i) \triangleq \exp\left(-\frac{C_{k:T}(X_{k:T}(i), U_{k:T-1}(i))}{\alpha}\right) \tag{7.68}$$

be the reward of the sample path i and $r \triangleq \sum_{i=1}^{N} r(i)$.

Introduce a piecewise constant, monotonically non-decreasing function $F : [0, N] \to [0, r]$ by

$$F(x) = \sum_{i=1}^{\lfloor x \rfloor} r(i)$$

The inverse F^{-1} of F defines a mapping $F^{-1} : [0, r] \to \{1, 2, ..., N\}$. To generate a random sample u_k, consider the following randomized algorithm: First, generate a random variable d according to $d \sim$ unif$[0, r]$. Second, select a sample ID by $i \leftarrow F^{-1}(d)$. Finally, select u_k as the control input adopted in the i-th sample path at time step k, i.e., $u_k \leftarrow U_k(i)$. The procedure is summarized in Algorithm 14.

Algorithm 14: Discrete-time continuous-state KL control by forward Monte-Carlo simulations

Given: Initial state x_0;
for $k = 0, 1, ..., T-1$ **do**
  Generate N sample paths $X_{k:t_f}(i), U_{k:t_f-1}(i), i = 1, 2, ..., N$ starting from x_k randomly according to the reference state transition policy $\{R_{U_k|X_k}\}_{t=k,...,T-1}$;
  For each $i = 1, 2, ..., N$, compute the path reward $r(i)$ by (7.68). Compute $r = \sum_{i=1}^{N} r(i)$.;
  Generate $d \sim$ unif$[0, r]$ randomly;
  Find an index $i^* \in \{1, 2, ..., N\}$ such that $\sum_{i=1}^{i^*-1} r(i) < d \leq \sum_{i=1}^{i^*} r(i)$;
  Select a control input as $u_k \leftarrow U_k(i^*)$;

The correctness of Algorithm 14 follows from Claim 6.1 in Section 6.4

7.3 Reinforcement learning revisited

In this section, we apply reinforcement learning (RL) algorithms to the class of KL control problems and see that their linearly solvable nature leads to a drastic simplification of the learning algorithms. As we saw in Chapter 3, RL algorithms can be categorized into the following three classes [4]:

- Model-based RL: This class of RL algorithms benefits from explicit knowledge about the environment, such as the cost function or the state transition kernels.
- Simulator-driven RL: This class of RL algorithms does not have access to explicit environment models, but a simulator is available to test the performance of a selected policy.
- Data-driven RL: This corresponds to the scenario where training data can only be obtained through the actual interaction with the environment.

Representative examples of these three classes are the value iteration (VI), Monte Carlo ES [5], and Q-learning [6] (Table 1). In this section, we will see that these algorithms reduce to linearly solvable VI, path integral control, and Z-learning, respectively, when they are applied to KL control problems (Table 1). These new algorithms are not only conceptually simpler but also computationally more efficient.

To receive these benefits, we need to make a few structural assumptions about the RL problems we consider, including the following:

1. The const function should be KL Divergence (KLD)-regularized;
2. The cost function should be undiscounted, i.e., $\gamma = 1$; and
3. The state transition should be deterministic, i.e., $x_{t+1} = F(x_t, u_t)$ for some function F.

	General control problems	KL control problems (This section)
Model-based RL	Value Iteration (VI)	Linearly solvable VI
Simulator-driven RL	Monte Carlo method	Path Integral Control
Data-driven RL	Q-learning	Z-learning

Table 7.1: Classification of RL algorithms.

Specifically, let us formulate the following KLD-regularized RL problem: As before, consider an MDP over a finite state space $\mathcal{X} = \{1, 2, \cdots, S, x_{\text{abs}}\}$ and a finite action space $\mathcal{U}$.[6] We assume that the state space contains the *absorbing state* $x^{\text{abs}} \in \mathcal{X}$, which is considered as the "end" of each episode.[7] Without loss of generality, we assume $C(x, u) > 0$ for all $x \in \mathcal{X} \setminus \{x^{\text{abs}}\}$ and $u \in \mathcal{U}$. The initial state is assumed to be $x_0 \in \mathcal{X} \setminus \{x^{\text{abs}}\}$. Let the random time T_f be the smallest time step t such that $x_t = x^{\text{abs}}$. We are interested in finding the optimal policy $\pi(u_t|x_t)$ that minimizes[8]

$$V^\mu(x) = \mathbb{E}^\mu \left[\sum_{t=0}^{T_f - 1} \left\{ C(x_t, u_t) + \alpha \log \frac{\mu(u_t|x_t)}{\mu'(u_t|x_t)} \right\} \Big| x_0 = x \right] \tag{7.69}$$

[6] For simplicity, we assume both $\mathcal{X}$ and $\mathcal{U}$ are finite spaces. However, the concepts for linearly solvable VI, path integral control, and Z-learning can be generalized to continuous spaces.

[7] We will use the absorbing state to ensure that episodes have finite lengths with probability one and the expected value of the undiscounted cost is finite.

[8] More generally, one can consider the sequence $\mu = (\mu_1, \mu_2, \cdots)$ of causal polices $\mu_t(u_t|x_{0:t}, u_{0:t-1})$. However, the domain of optimization can be restricted to the class of time-invariant and Markov policies $\mu(u_t|x_t)$ without loss of generality.

simultaneously for all $x \in \mathcal{X} \setminus \{x^{\text{abs}}\}$. We set $V^\mu(x^{\text{abs}}) = 0$ for all μ, assuming that the absorbing state incurs no cost. With this setup, (7.69) can also be viewed as an infinite-horizon stochastic optimal control problem in which the stage-wise cost is zero for all $t \geq T_f$. In (7.69), α is a positive constant, $\mu'(u_t|x_t)$ is a fixed (given) policy, and

$$\mathbb{E}^\mu \log \frac{\mu(u_t|x_t)}{\mu'(u_t|x_t)} = \sum_{u_t} \mu(u_t|x_t) \frac{\mu(u_t|x_t)}{\mu'(u_t|x_t)} \tag{7.70}$$

$$= D(\mu(\cdot|x_t) \| \mu'(\cdot|x_t)) \tag{7.71}$$

is the KL divergence.

7.3.1 Model-based RL: Linearly solvable value iteration

Let us begin with model-based synthesis, assuming that explicit models of F, C, and π' are available. By Bellman's optimality principle, the value function satisfies

$$V^*(x_t) = \min_{\mu(u_t|x_t)} \sum_{u_t} \mu(u_t|x_t) \left\{ \rho(x_t, u_t) + \alpha \log \frac{\mu(u_t|x_t)}{\mu'(u_t|x_t)} \right\} \tag{7.72}$$

where $\rho(x_t, u_t) = C(x_t, u_t) + V^*(F(x_t, u_t))$. This equation is identical to (7.18), except that the optimal value function V^* is already assumed to be time-invariant. This is justified by the time-invariant and infinite-horizon nature of the problem setup (7.69). Note that the optimal policy is available and is given as the Boltzmann distribution:

$$\mu^*(u_t|x_t) = \frac{\mu'(u_t|x_t) \exp\{-\rho(x_t, u_t)/\alpha\}}{\sum_{u_t'} \mu'(u_t'|x_t) \exp\{-\rho(x_t, u_t')/\alpha\}}. \tag{7.73}$$

Substituting (7.73) back to (7.72) simplifies it to

$$V^*(x_t) = -\alpha \log \sum_{u_t} \mu'(u_t|x_t) \exp\left\{ -\frac{C(x_t, u_t)}{\alpha} - \frac{V^*(x_{t+1})}{\alpha} \right\}. \tag{7.74}$$

Introducing the *desirability function* $Z^*(x) := \exp(-V^*(x)/\alpha)$, this equation can be further simplified to

$$Z^*(x_t) = \sum_{u_t} \mu'(u_t|x_t) \exp\left(-\frac{C(x_t, u_t)}{\alpha} \right) Z^*(x_{t+1}) \tag{7.75}$$

for $x_t \in \mathcal{X} \setminus \{x_{\text{abs}}\}$ and $Z^*(x_{\text{abs}}) = \exp(-V^*(x_{\text{abs}})/\alpha) = 1$. As we have already observed in (7.25), this is a linear equation of Z^*. The only difference from (7.25) is that Z^* in (7.75) is time-invariant, and that a special treatment is needed to the absorbing state. To be more explicit, let us index the state space as $\mathcal{X} = \{1, 2, \cdots, S, x_{\text{abs}}\}$, and let

$$\bar{Z}^* = \begin{bmatrix} Z^*(1) & Z^*(2) & \cdots & Z^*(S) \end{bmatrix}^\top \in \mathbb{R}^S$$

be the vector of desirability values. (We have omitted $Z^*(x_{\text{abs}}) = 1$ from the list). Equation (7.75) implies that $\bar{Z}^*$ satisfies

$$\begin{bmatrix} Z^*(1) \\ \vdots \\ Z^*(S) \end{bmatrix} = \underbrace{\begin{bmatrix} a_{11} & \cdots & a_{1S} \\ \vdots & & \vdots \\ a_{S1} & \cdots & a_{SS} \end{bmatrix}}_{A} \begin{bmatrix} Z^*(1) \\ \vdots \\ Z^*(S) \end{bmatrix} + \underbrace{\begin{bmatrix} w_1 \\ \vdots \\ w_S \end{bmatrix}}_{w}.$$

Entries of A and w are

$$a_{ij} = \sum\nolimits_{u \in \mathcal{U}_{i \to j}} \mu'(u|i) \exp\left(-C(i,u)/\alpha\right)$$
$$w_i = \sum\nolimits_{u \in \mathcal{U}_{i \to x_{\text{abs}}}} \mu'(u|i) \exp\left(-C(i,u)/\alpha\right)$$

where $\mathcal{U}_{i \to j} = \left\{u \in \mathcal{U} \middle| F(i,u) = j\right\}$. If the matrix A is contractive, a simple recursion

$$\bar{Z}_{k+1} = A\bar{Z}_k + w, \;\; \bar{Z}_0 > 0 \text{ (arbitrary)} \tag{7.76}$$

achieves convergence $\bar{Z}_k \to \bar{Z}^*$ as $k \to \infty$. This implies that, unlike the generic value iteration (which involved a "min" operator), the value iteration (7.76) only requires additions and multiplications.

Exercise 7.8 *Show that matrix A is contractive if $C(x,u) > 0$ for all $x \in \mathcal{X} \setminus \{x_{abs}\}$ and $u \in \mathcal{U}$. Hint: Use the Perron-Frobenius theorem.*

Exercise 7.9 *Instead of* (7.69)*, consider the problem with a discounted cost function:*

$$V^{\mu}(x) = \mathbb{E}^{\mu}\left[\sum_{t=0}^{T_f-1} \gamma^t \left\{C(x_t,u_t) + \alpha \log \frac{\mu(u_t|x_t)}{\mu'(u_t|x_t)}\right\} \middle| x_0 = x\right] \tag{7.77}$$

where $\gamma \in (0,1)$ is a discount factor. Is (7.77) *linearly solvable?*

7.3.2 Simulator-driven RL

Next, let us consider a scenario where the models F, C, and μ' are not available but episodes $x_0, u_0, x_1, u_1, \cdots, x_{T_f}$ under the nominal policy μ' are easy to simulate. As we saw in Chapter 3, Monte Carlo methods are useful in such scenarios.

Equation (7.75) implies that the desirability function at time step t can be expressed in terms of the desirability function at time step $t+1$. Since the same argument holds between time steps $t+1$ and $t+2$, $Z^*(x_t)$ can also be expressed as

$$\begin{aligned} Z^*(x_t) = &\sum_{u_t} \mu'(u_t|x_t) \exp\left(-\frac{C(x_t,u_t)}{\alpha}\right) \\ &\times \sum_{u_{t+1}} \mu'(u_{t+1}|x_{t+1}) \exp\left(-\frac{C(x_{t+1},u_{t+1})}{\alpha}\right) Z^*(x_{t+2}) \end{aligned}$$

where $x_{t+1} = F(x_t,u_t)$ and $x_{t+2} = F(x_{t+1},u_{t+1})$. By repeating this argument from $t=0$ till the end of the episode, we obtain

$$\begin{aligned} Z^*(x_0) &= \sum_{u_{0:T_f-1}} \prod_{t=0}^{T_f-1} \mu'(u_t|x_t) \exp\left\{-\frac{1}{\alpha}\sum_{t=0}^{T_f-1} C(x_t,u_t)\right\} \\ &= \mathbb{E}^{\mu'} \exp\left\{-\frac{1}{\alpha}\sum_{t=0}^{T_f-1} C(x_t,u_t)\right\} \end{aligned} \tag{7.78}$$

where $x_{t+1} = F(x_t, u_t)$ for $t = 0, 1, \cdots, T_f - 1$.

The result (7.78) indicates that $Z^*(x_0)$ can be computed by Monte Carlo simulations of full episodes. Specifically, for $i = 1, 2, \cdots, N$, let

$$x_0, u_0^i, x_1^i, u_1^i, \cdots, x_{T_f-1}^i, u_{T_f-1}^i$$

be N independently simulated episodes starting from x_0 under the nominal policy μ'. By the law of large numbers, we expect that the empirical mean converges to $Z^*(x_0)$ quadratically as N tends to infinity:

$$Z^*(x_0) \approx \frac{1}{N} \sum_{i=1}^{N} \exp\left\{ -\frac{1}{\alpha} \sum_{t=0}^{T_f-1} C(x_t^i, u_t^i) \right\}. \tag{7.79}$$

Consequently, the value function $V^*(x)$ of the KLD-regularized RL problem (7.69) can also be computed by Monte Carlo simulations as follows:

$$V^*(x_0) \approx -\alpha \log \left[\frac{1}{N} \sum_{i=1}^{N} \exp\left\{ -\frac{1}{\alpha} \sum_{t=0}^{T_f-1} C(x_t^i, u_t^i) \right\} \right]. \tag{7.80}$$

This formulation highlights a key insight: the value function can be estimated directly from simulated path costs (path integrals) generated under the nominal policy. This fundamental idea underpins the path integral control algorithms, which we will study further in the next chapter.

Notice also that by generating sample trajectories $x_0, u_0, x_1^i, u_1^i, \cdots, x_{T_f-1}^i, u_{T_f-1}^i$ starting from a particular state-control pair (x_0, u_0), one can approximate the optimal Q-function Q^* by

$$Q^*(x_0, u_0) \approx -\alpha \log \left[\frac{1}{N} \sum_{i=1}^{N} \exp\left\{ -\frac{1}{\alpha} C(x_0, u_0) - \frac{1}{\alpha} \sum_{t=1}^{T_f-1} C(x_t^i, u_t^i) \right\} \right]. \tag{7.81}$$

A major distinction exists between (7.81) and the Monte Carlo methods (Algorithms 7 and 8) employed for generic RL. As evident from Algorithms 7 and 8, Monte-Carlo-based RL in the generic setup requires a simultaneous search for both the optimal Q-function Q^* and the policy μ^*, which is achieved via policy iteration. In contrast, the approach in (7.81) circumvents the need for explicit knowledge of the optimal policy μ^*. Since episodes can be generated directly using the known nominal policy μ', the Q-function in (7.81) can be computed from a single batch of Monte Carlo simulations. This simplifies the overall algorithm substantially. Additionally, *importance sampling* [4, 7] can be employed to estimate (7.78) using samples generated by an alternative behavior policy $\mu''(\neq \mu')$.

7.3.3 Data-driven RL: Z-learning

In Chapter 3, we applied the concept of TD learning to the Monte Carlo method (Section 3.3) and obtained Q-learning (Algorithm 9). In a similar vein, we can apply TD learning to the Monte Carlo method (7.80) and obtain the so-called Z-learning algorithm [3].

Like Q-learning, Z-learning allows the estimate $\hat{Z}$ of the desirability function to be updated using data (physical or simulated) that becomes available in each time step. Let x_t, u_t, x_{t+1} be the transition generated by the behavioral policy μ', and let $C_t = C(x_t, u_t)$ be the observed cost. The estimate $\hat{Z}$ is updated online as

$$\hat{Z}_{t+1}(x_t) = (1 - \theta_t)\hat{Z}_t(x_t) + \theta_t \exp\left(-\frac{C_t}{\alpha}\right)\hat{Z}_t(x_{t+1}) \tag{7.82}$$

where θ_t is a diminishing learning rate.

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